Homework Assignment: NumPy, Pandas, and Matplotlib

NumPy Tasks
Task 1: Create a NumPy array of integers from 1 to 20. Compute the mean, maximum, and standard deviation.
Hint: Look at NumPy statistical functions.
Must Use: numpy.mean(), numpy.max(), numpy.std()

Task 2: Create a 5×5 random matrix and replace all values less than 0.5 with 0.
Hint: Boolean masking is useful here.
Must Use: numpy.random.rand(), boolean indexing

Pandas Tasks
Task 3: Load a CSV file, display the first 5 rows, and calculate the average of one numeric column.
Hint: Use DataFrame summary functions.
Must Use: pandas.read_csv(), DataFrame.head(), DataFrame.mean()

Task 4: Filter rows where a column value exceeds a threshold and save the result to a new CSV file.
Hint: Boolean conditions help in filtering.
Must Use: DataFrame.loc[], DataFrame.to_csv()

Matplotlib Tasks
Task 5: Plot a line graph of values from 1 to 10 and label axes properly.
Hint: Labels and titles improve readability.
Must Use: matplotlib.pyplot.plot(), xlabel(), ylabel(), title()

Task 6: Create a bar chart showing values for 5 categories.
Hint: Bar plots require categorical labels.
Must Use: matplotlib.pyplot.bar(), xticks()